DeRO: Dead Reckoning Based on Radar Odometry With Accelerometers Aided for Robot Localization



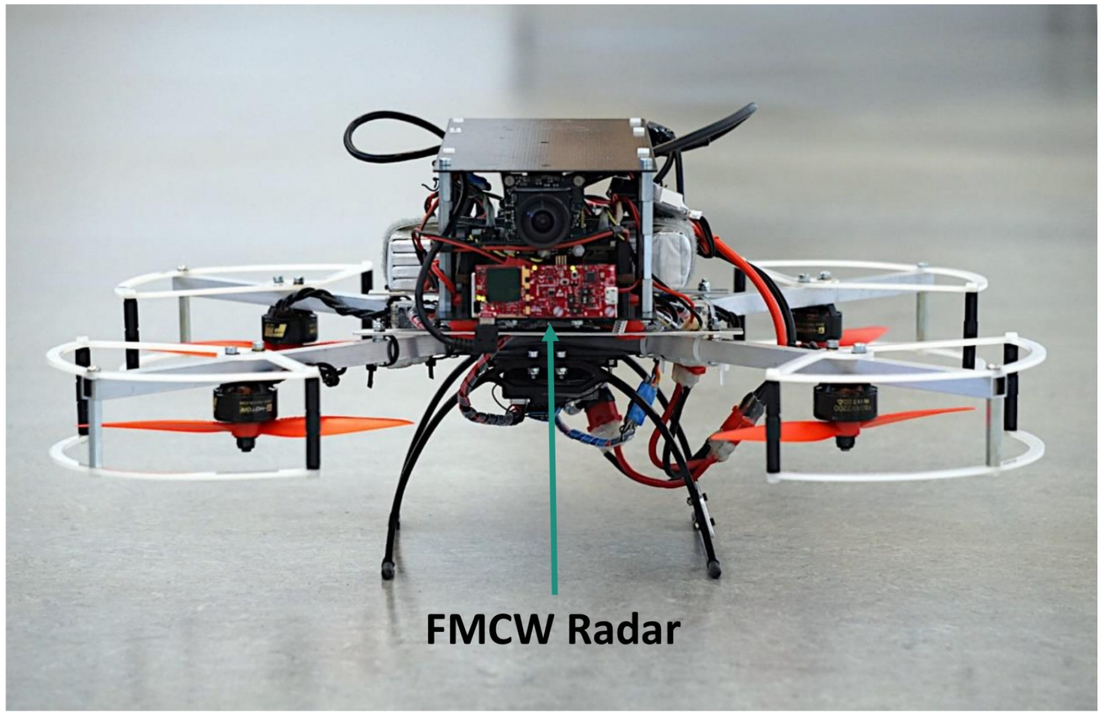
Hoang Viet Do, Yong Hun Kim,
Joo Han Lee, Min Ho Lee, and Jin Woo Song*

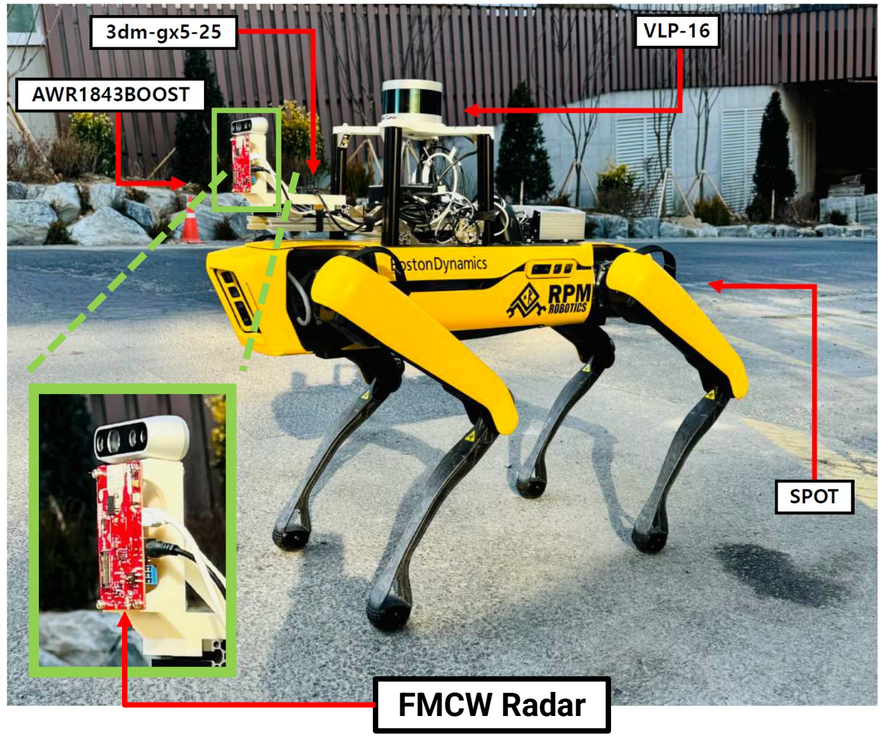
Sejong University, Seoul, Republic Of Korea

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Introduction (1/3)

- mmWave FMCW Radar has gained significant attention.
  - Low cost, compact size, and all-weather operation.



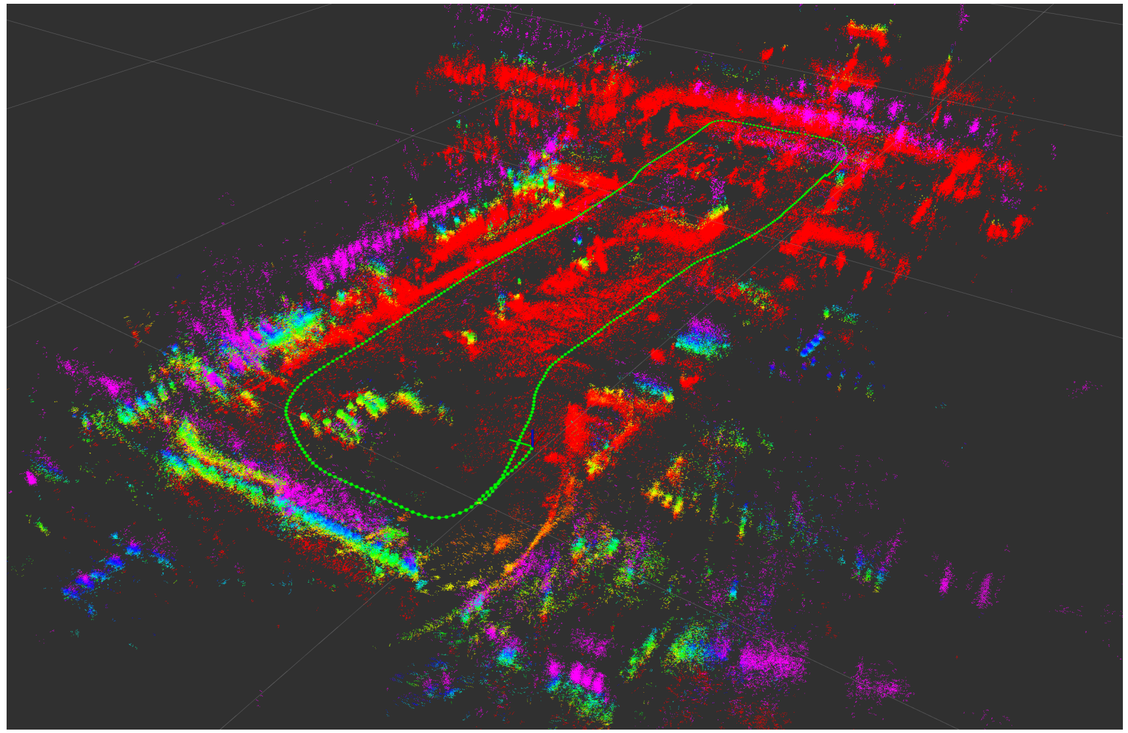


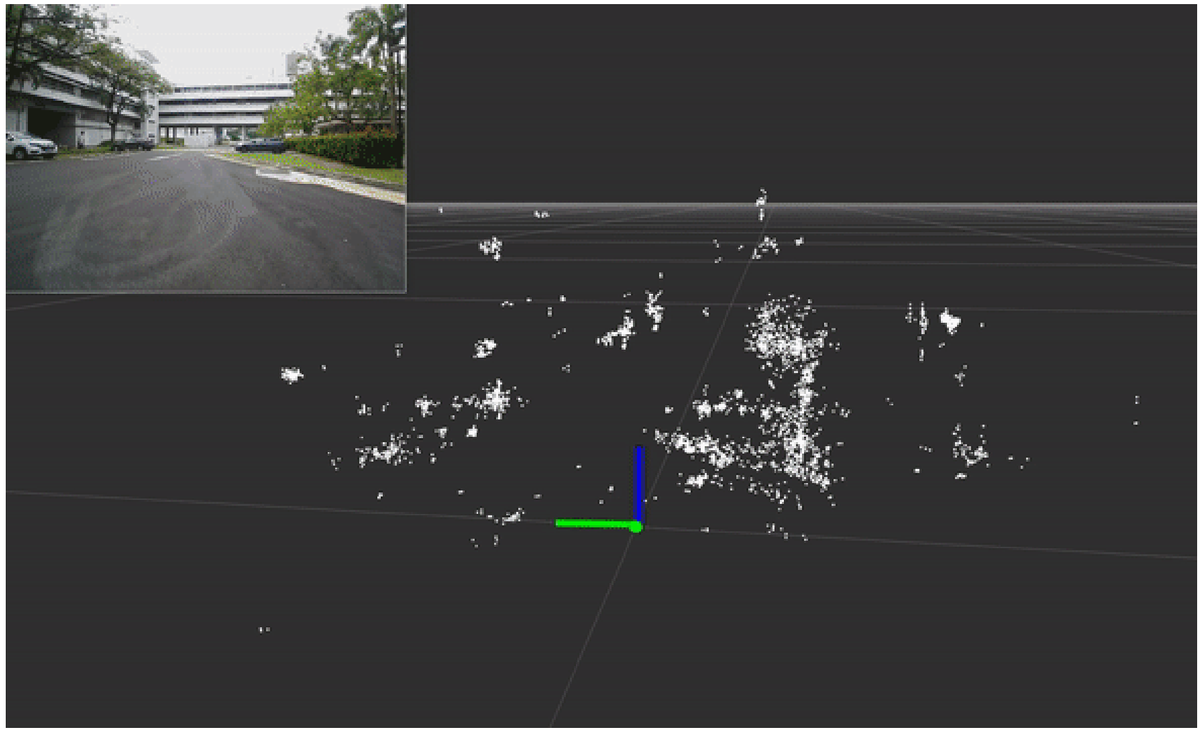
C. Doer and G. F. Trommer, "Radar Inertial Odometry With Online Calibration," *2020 European Navigation Conference (ENC)*, Dresden, Germany, 2020, pp. 1-10, doi: 10.23919/ENC48637.2020.9317343.

Jung, Sangwoo, Wooseong Yang, and Ayoung Kim. "Co-RaL: Complementary Radar-Leg Odometry with 4-DoF Optimization and Rolling Contact." *arXiv preprint arXiv:2407.05820 (2024)*.

Introduction (2/3)

- Promising alternative to LiDAR and Camera for SLAM.
  - LiDAR and Camera are often limited by poor lighting, illumination, and environment factors.

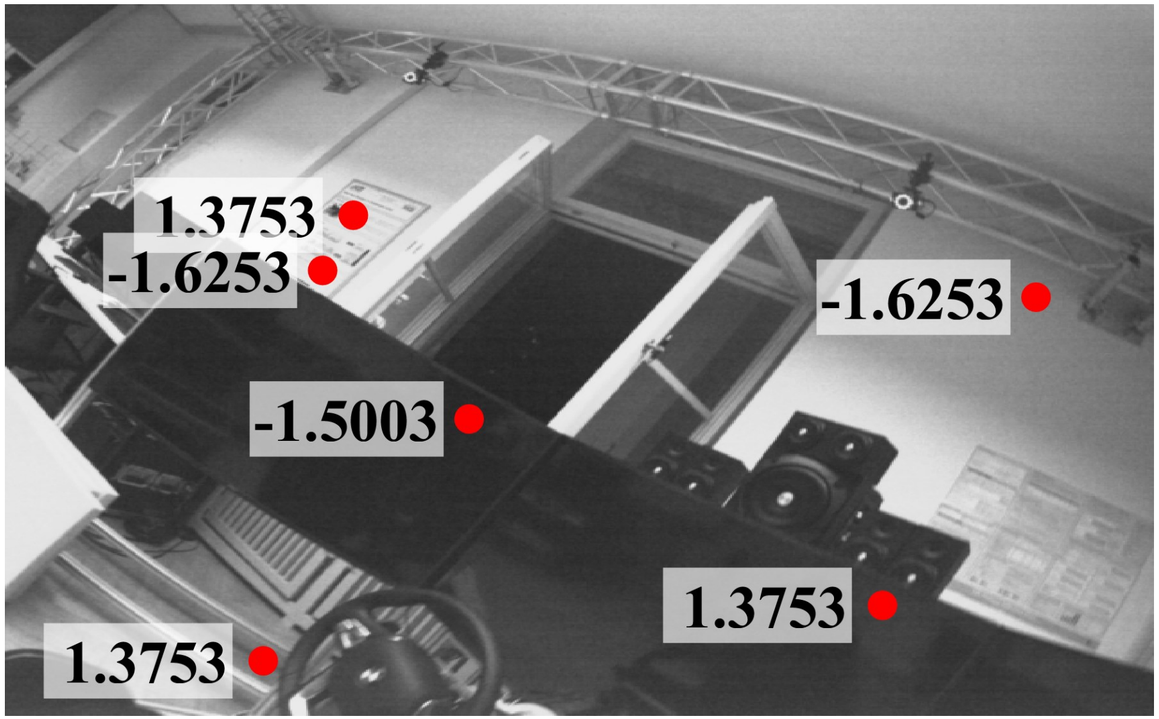


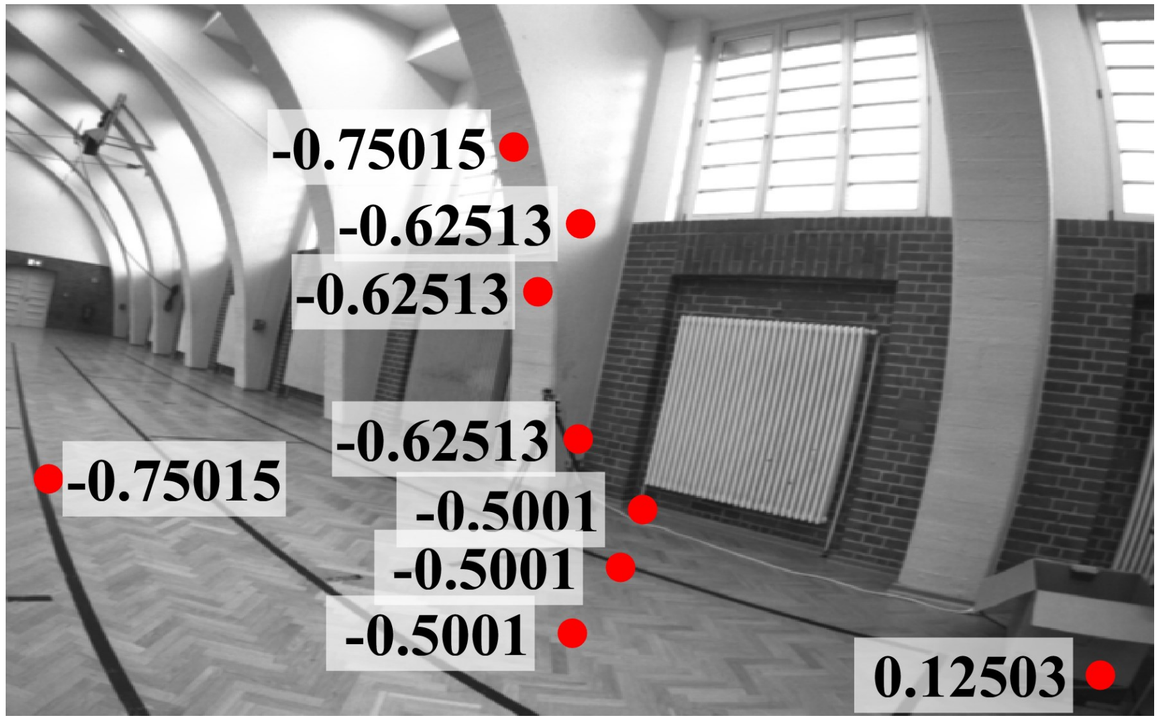


J. Zhang, H. Zhuge, Z. Wu, G. Peng, M. Wen, Y. Liu, and D. Wang, “4dradarslam: A 4d imaging radar slam system for large-scale environments based on pose graph optimization,” *in 2023 IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 8333-8340.

Introduction (3/3)

- Radar point clouds are known to be noisy and sparse due to ghost targets and multipath signals.
  - Typical commercial radar provides 4D point cloud (range, bearing and Doppler velocity).

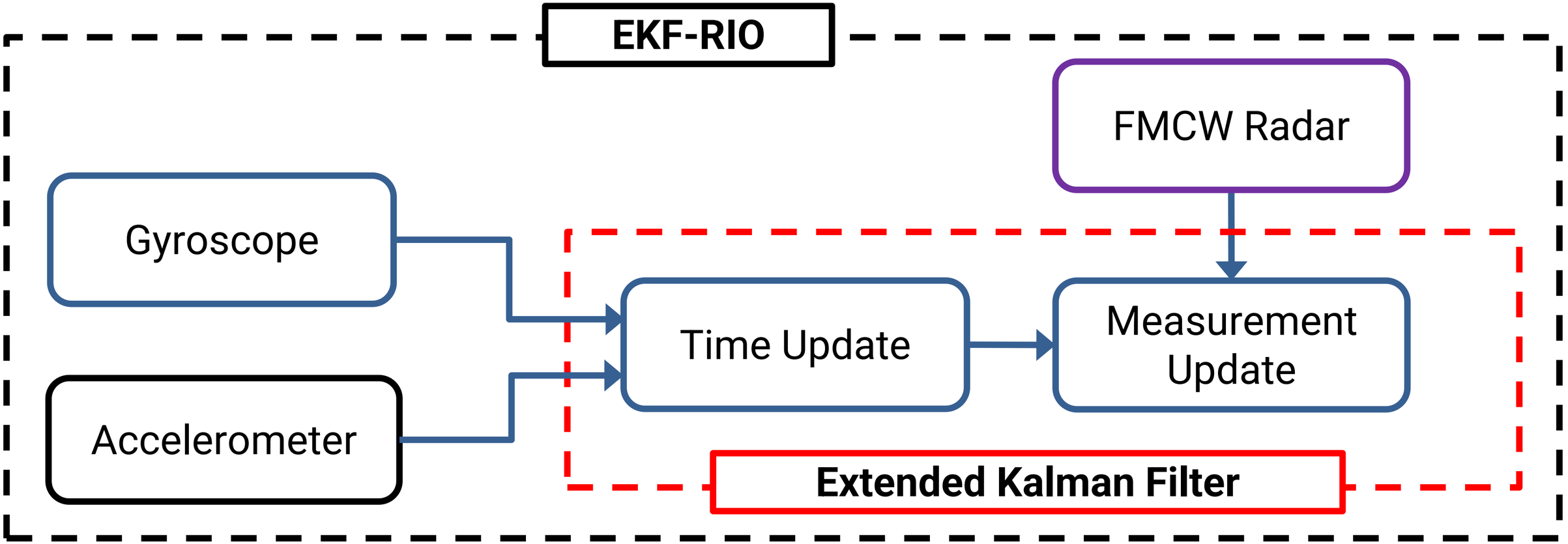




Hoang Viet Do, Bo Sung Ko, and Jin Woo Song. "CREVE: An Acceleration-based Constraint Approach for Robust Radar Ego-Velocity Estimation." *arXiv preprint arXiv:2409.16847* (2024).

Traditional Approach

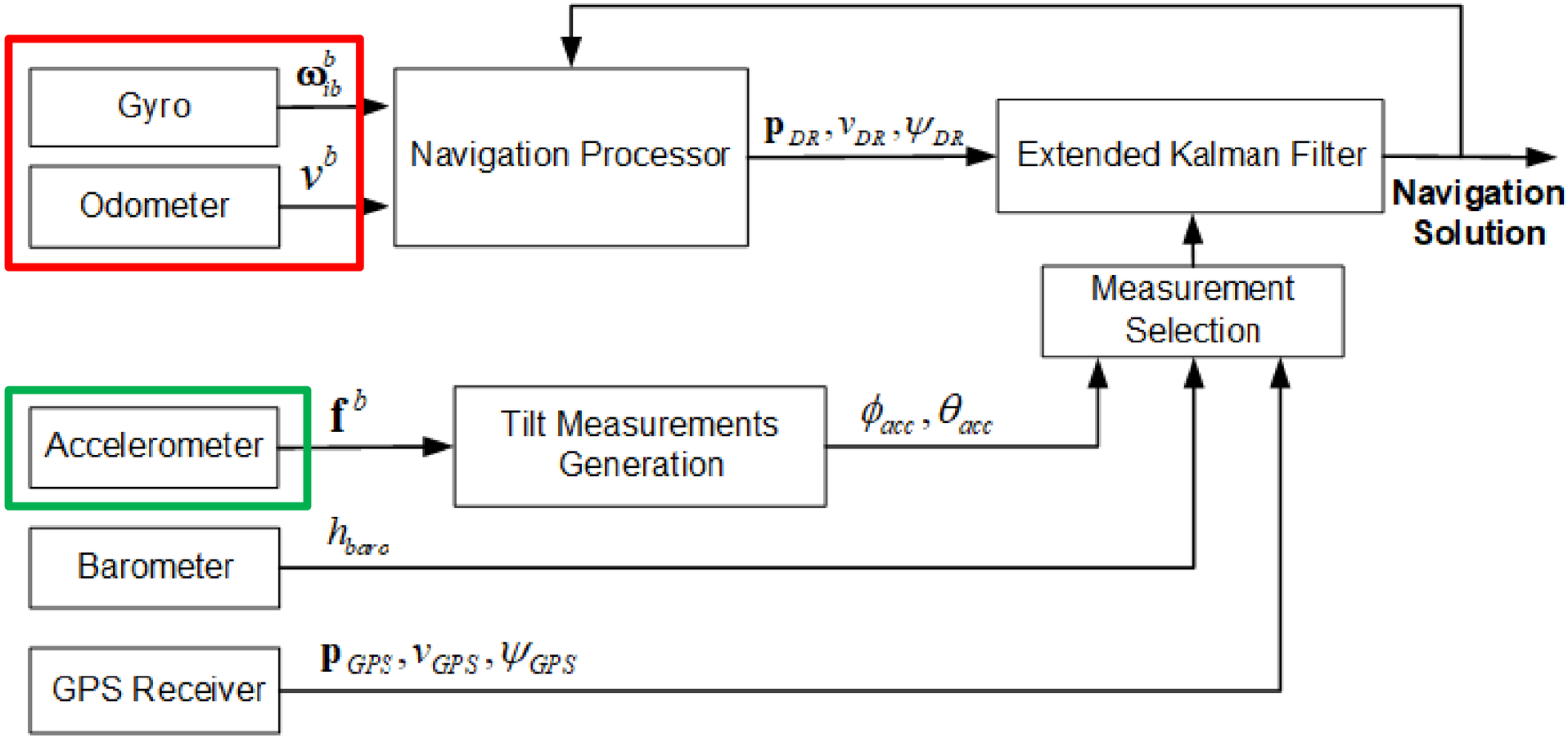
- Extended Kalman Filter - Radar Inertial Odometry (EKF-RIO).
  - Radar Doppler velocity is used as an auxiliary measurement for error correction.



C. Doer and G. F. Trommer, “An ekf based approach to radar inertial odometry,” in *2020 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, 2020, pp. 152–159.

Dead Reckoning (1/2)

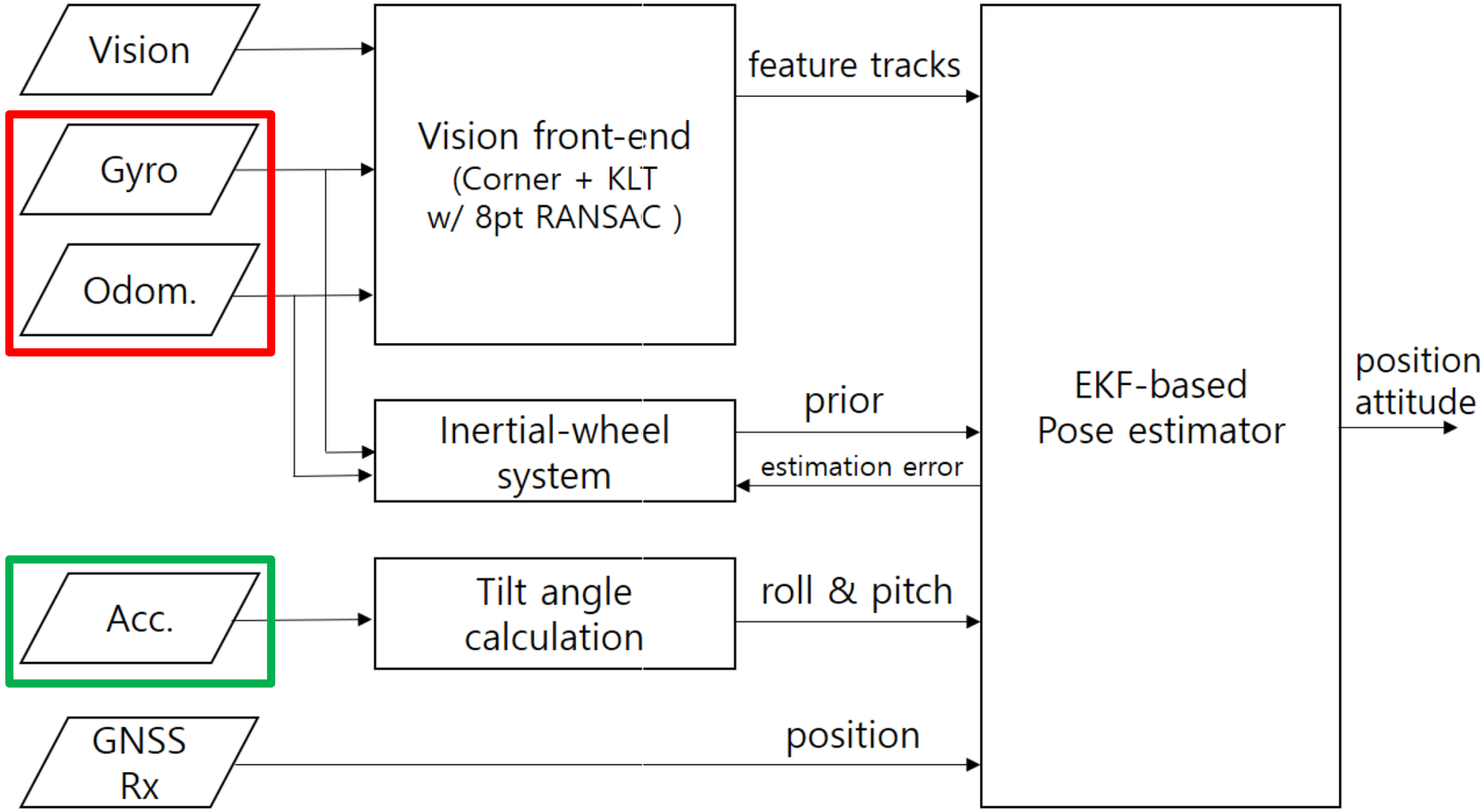
- Dead Reckoning (DR) is widely-used solution for ground vehicles.
  - Requires velocity sensor (e.g., odometer) to replace the accelerometer.



W. J. Park, J. W. Song, C. H. Kang, J. H. Lee, M. H. Seo, S. Y. Park, J. Y. Yeo, and C. G. Park, “Mems 3d dr/gps integrated system for land vehicle application robust to gps outages,” *IEEE Access*, vol. 7, pp. 73 336–73 348, 2019.

Dead Reckoning (2/2)

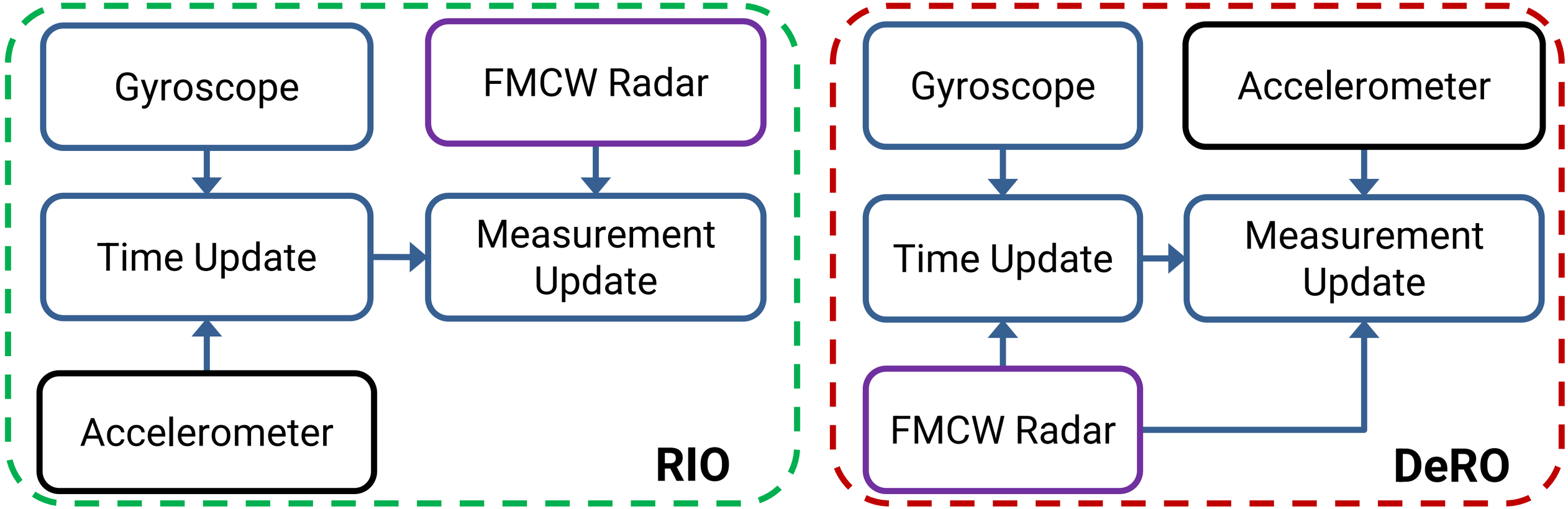
- DR technique is also used in vision-based navigation system.



J. H. Jung, J. Cha, J. Y. Chung, T. I. Kim, M. H. Seo, S. Y. Park, J. Y. Yeo, and C. G. Park, "Monocular visual-inertial-wheel odometry using low-grade imu in urban areas," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 2, pp. 925–938. 2022.

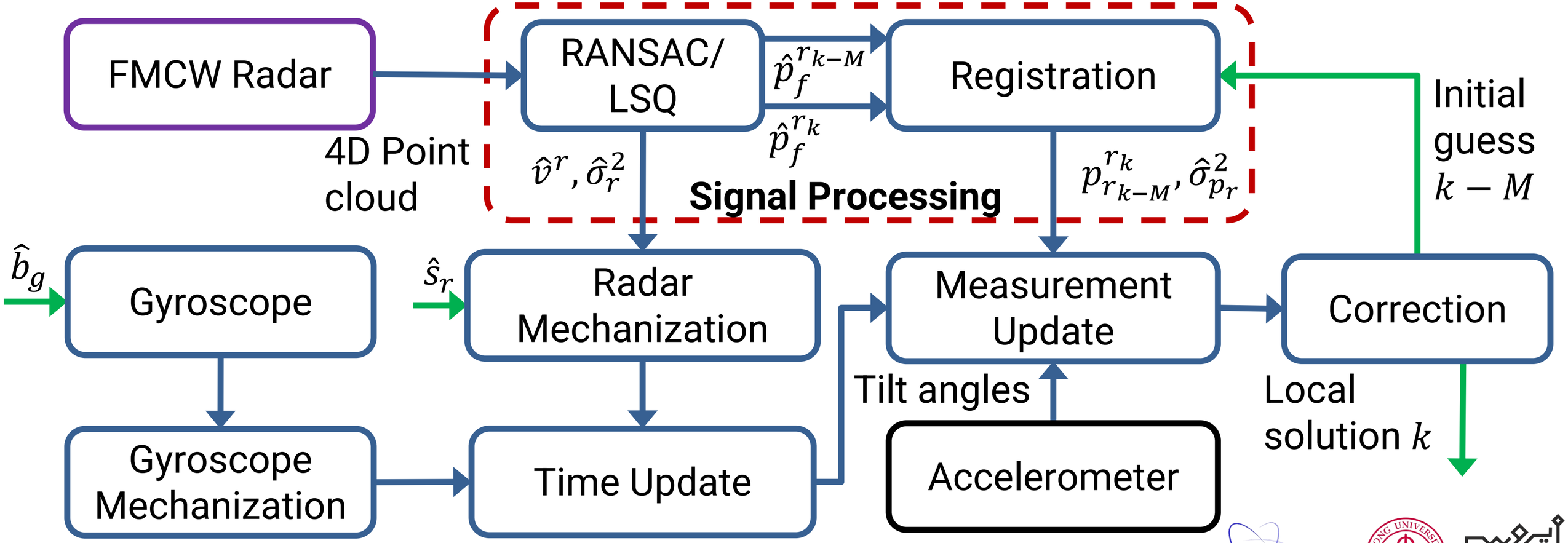
Key Takeaways

- Apply DR technique to radar-inertial navigation systems.
  - Use radar measurements for time propagation and error correction.



DeRO (1/6) – Overview

- Radar Doppler velocity is used to calculate position (radar mechanization).
- Radar range and bearing are used for error correction.



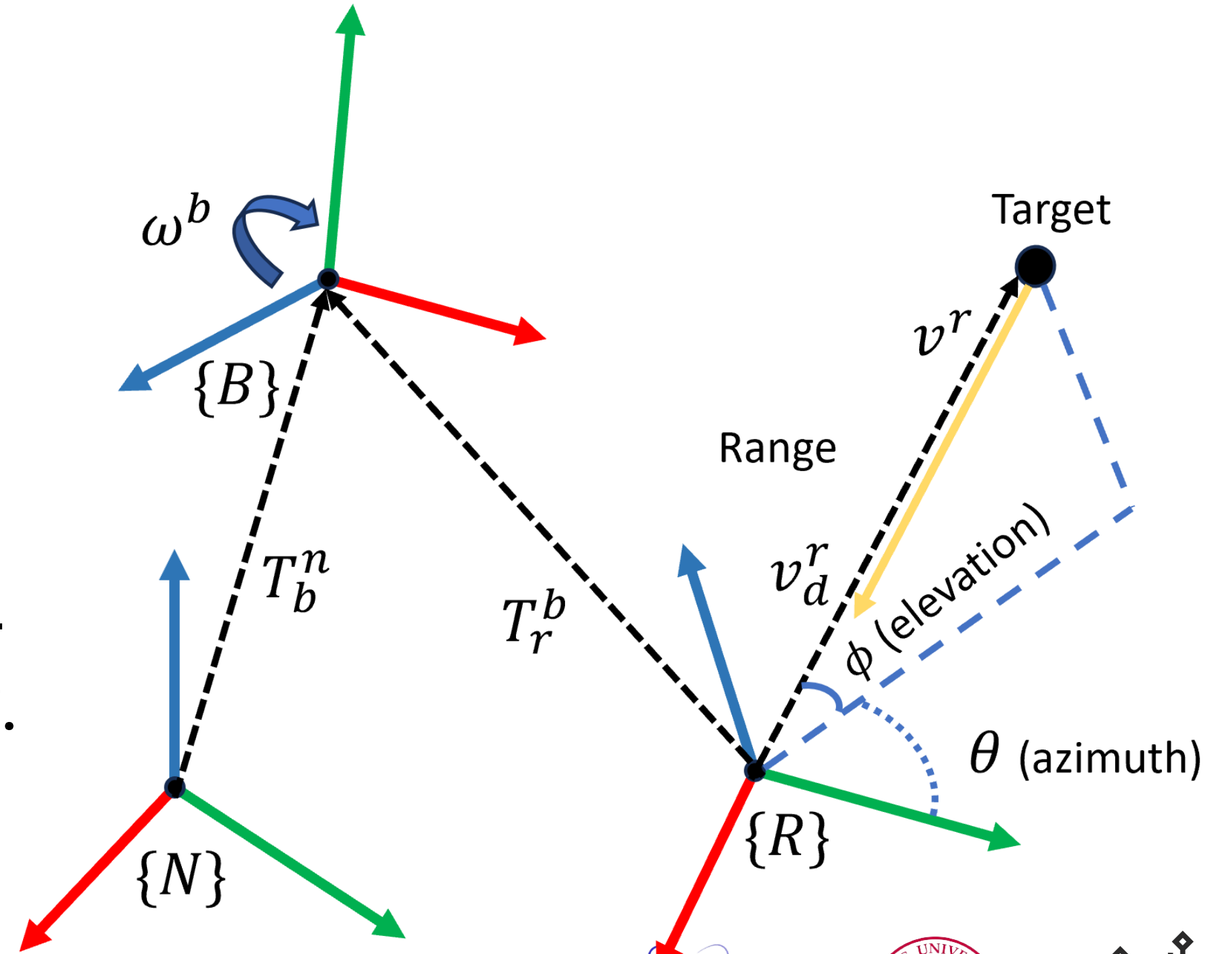
DeRO (2/6) – Radar Ego-Velocity Estimation

- Given a set of at least three point clouds $P_i = \begin{bmatrix} p^r_{x,i} & p^r_{y,i} & p^r_{z,i} & v_{D,i} \end{bmatrix}$.
  - Robust 3D radar ego-velocity can be estimated using RANSAC/LSQ.

$$\begin{bmatrix} -v_{D,1} \\ -v_{D,1} \\ \vdots \\ -v_{D,N} \end{bmatrix} = \begin{bmatrix} p^r_{x,1} & p^r_{y,1} & p^r_{z,1} \\ p^r_{x,2} & p^r_{y,2} & p^r_{z,2} \\ \vdots & \vdots & \vdots \\ p^r_{x,N} & p^r_{y,N} & p^r_{z,N} \end{bmatrix} \begin{bmatrix} v^r_x \\ v^r_y \\ v^r_z \end{bmatrix}$$

- Relationship between radar ego-velocity and body frame velocity.

$$v^r = C^r_b \left(C^b_n v^n_b + [\omega^b]_\times p^b_r \right)$$



DeRO (3/6) – Mechanization

- DeRO: Integrate radar velocity **once** to obtain position.
  - Requires radar-IMU extrinsic parameters.
- RIO: Integrate acceleration **twice** to obtain position.
  - Double integration.
  - Requires gravitational force and accelerometer bias estimation.

$$\begin{cases} \dot{p}_b^n = v_b^n \\ \dot{v}_b^n = C_b^n f^b + g^n \\ \dot{q}_b^n = \frac{1}{2}[q_b^n]_L \begin{bmatrix} 0 \\ \omega^b \end{bmatrix} \end{cases}$$

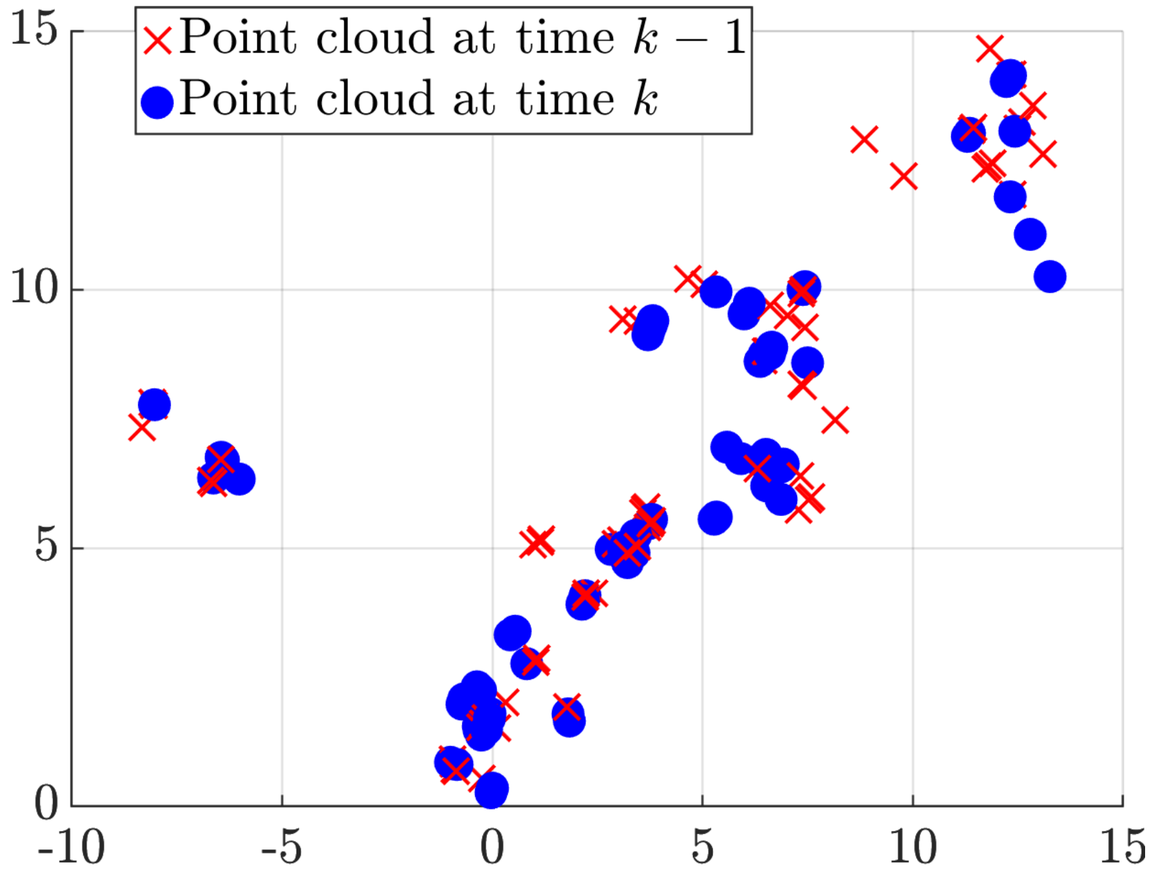
RIO

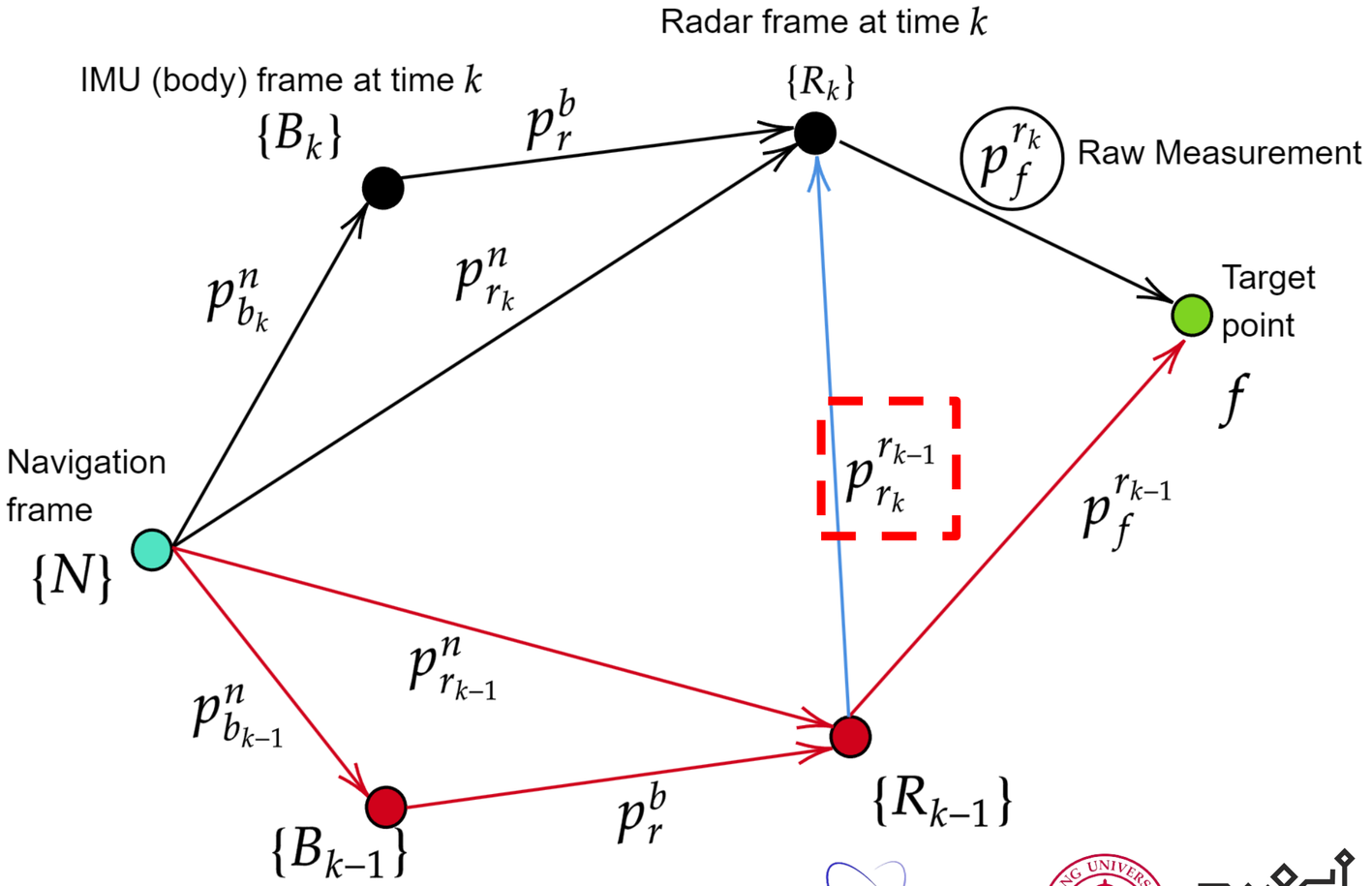
$$\begin{cases} \dot{p}_b^n = C_b^n \left(C_b^r v^r - [\omega^b]_\times p_r^b \right) \\ \dot{q}_b^n = \frac{1}{2}[q_b^n]_L \begin{bmatrix} 0 \\ \omega^b \end{bmatrix} \end{cases}$$

DeRO

DeRO (4/6) – Point Cloud Registration

- The relative distance between two radar scans $p_{r_k}^{r_{k-1}}$ can be estimated using point cloud registration.





DeRO (5/6) – Stochastic Cloning EKF

- Cloning M-step previous estimated state variable.
- Two measurements:
  - Relative distance and tilt angle.

Cloning state variable

$$x = \begin{bmatrix} p_{b,k}^{n\top} & v_{b,k}^{n\top} & q_{b,k}^{n\top} & b_{a,k}^{\top} & b_{g,k}^{\top} \end{bmatrix}^{\top}$$

$$y = \hat{v}^{r} - v^{r}$$

RIO

$$x = \begin{bmatrix} x_{DR}^{\top} & p_{b,k-M}^{n\top} & q_{b,k-M}^{n\top} \end{bmatrix}^{\top}$$

$$x_{DR} = \begin{bmatrix} p_{b,k}^{n\top} & q_{b,k}^{n\top} & b_{g,k}^{\top} & s_{r,k}^{\top} \end{bmatrix}^{\top}$$

$$y = \hat{p}_{r_k}^{r_{k-M}} - p_{r_k}^{r_{k-M}}$$

$$y_a = \begin{bmatrix} \hat{\phi} \\ \hat{\theta} \end{bmatrix} - \begin{bmatrix} \phi_{accel} \\ \theta_{accel} \end{bmatrix}$$

DeRO

DeRO (6/6) – Tilt Angles

- Use radar ego-velocity estimation to compensate for the accelerometer's acceleration for accurate tilt angle calculation.

$$\hat{f}^b = f^b - \hat{b}_a - \hat{C}_n^b \frac{(\hat{v}_{b,k}^n - \hat{v}_{b,k-1}^n)}{T}$$

- Accelerometer bias is obtained from the pre-calibration process.
- For further robust estimation, we use an adaptive strategy to adjust the tilt angle noise covariance.

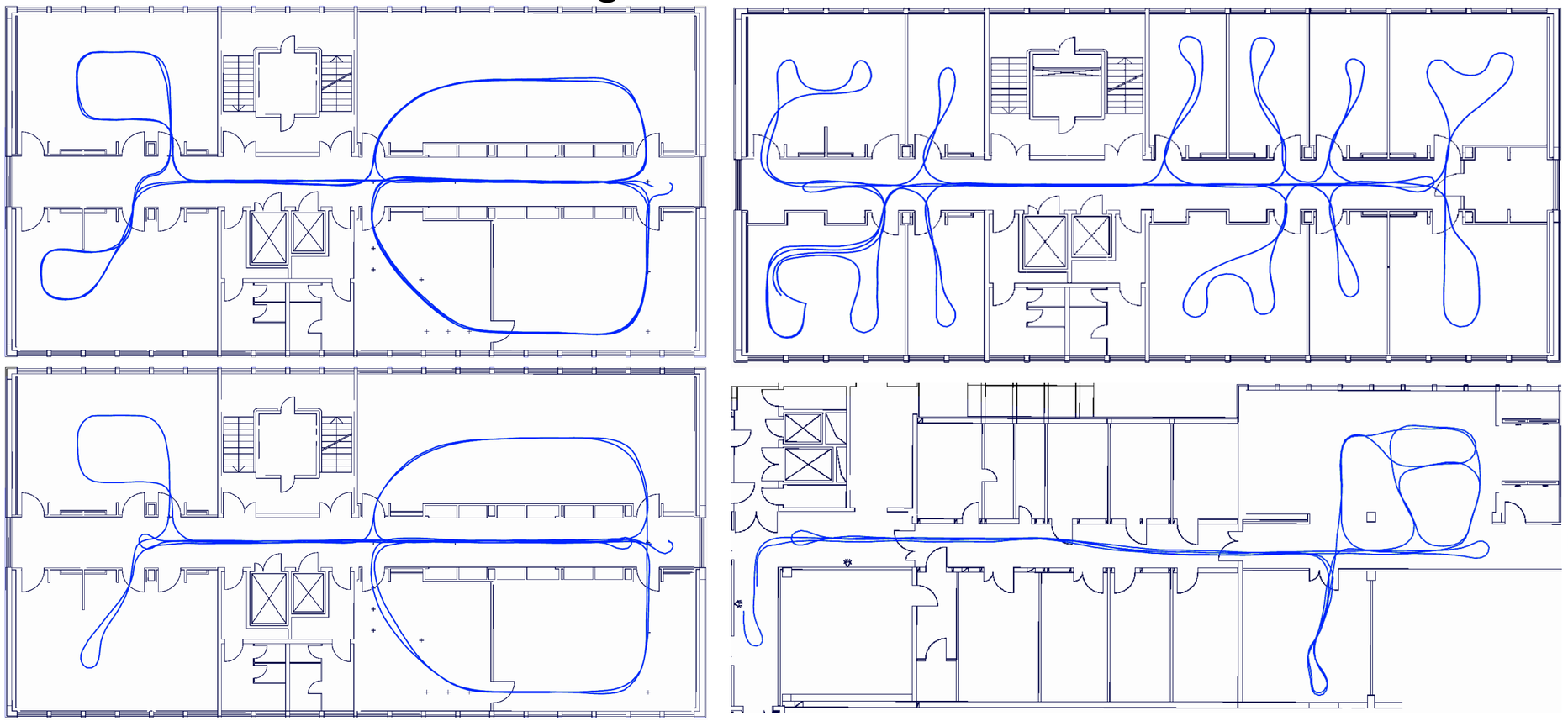
$$\begin{cases} \uparrow R_{accel} & if \ \ \left\|\hat{f}^b\right\| - \|g\| > \gamma \\ \downarrow R_{accel} & otherwise \end{cases}$$

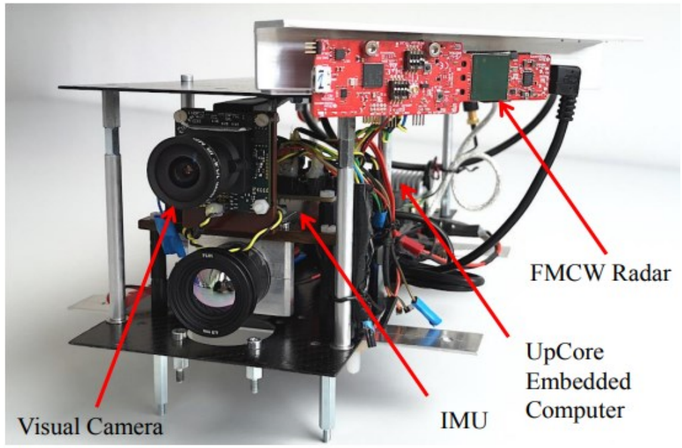
W. J. Park, J. W. Song, C. H. Kang, J. H. Lee, M. H. Seo, S. Y. Park, J. Y. Yeo, and C. G. Park, "Mems 3d dr/gps integrated system for land vehicle application robust to gps outages," *IEEE Access*, vol. 7, pp. 73 336–73 348, 2019.

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Experiments (1/6)

- To verify the proposed algorithm, 5 open-source datasets is used.
  - 4D mmWave FMCW radar – TI IWR6843AOP – 10 Hz.
  - IMU – Analog Devices ADIS16448 – 400 Hz.



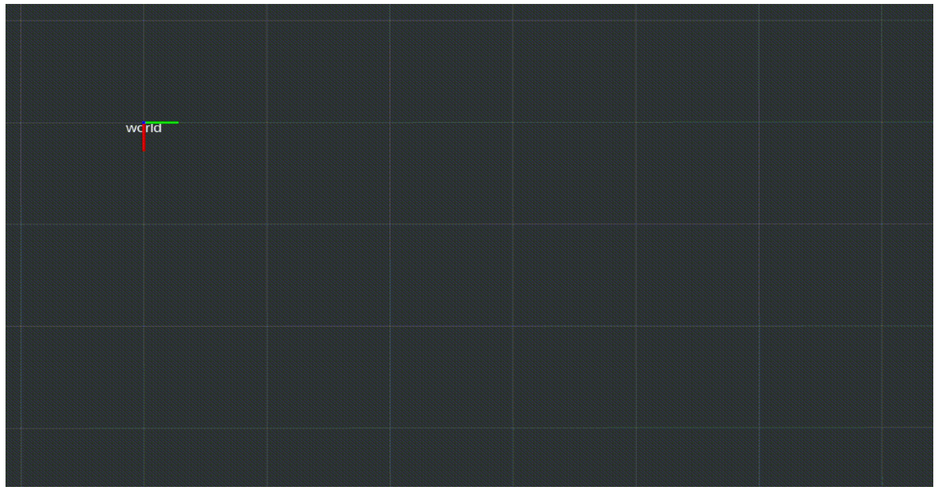


C. Doer and G. F. Trommer, "Yaw aided radar inertial odometry using manhattan world assumptions," in 2021 28th Saint Petersburg International Conference on Integrated Navigation Systems (ICINS), 2021, pp. 1–9.

Experiments (2/6)

- DeRO is implemented in ROS 2 Foxy using C++.
- We have open-sourced our implementation on GitHub.





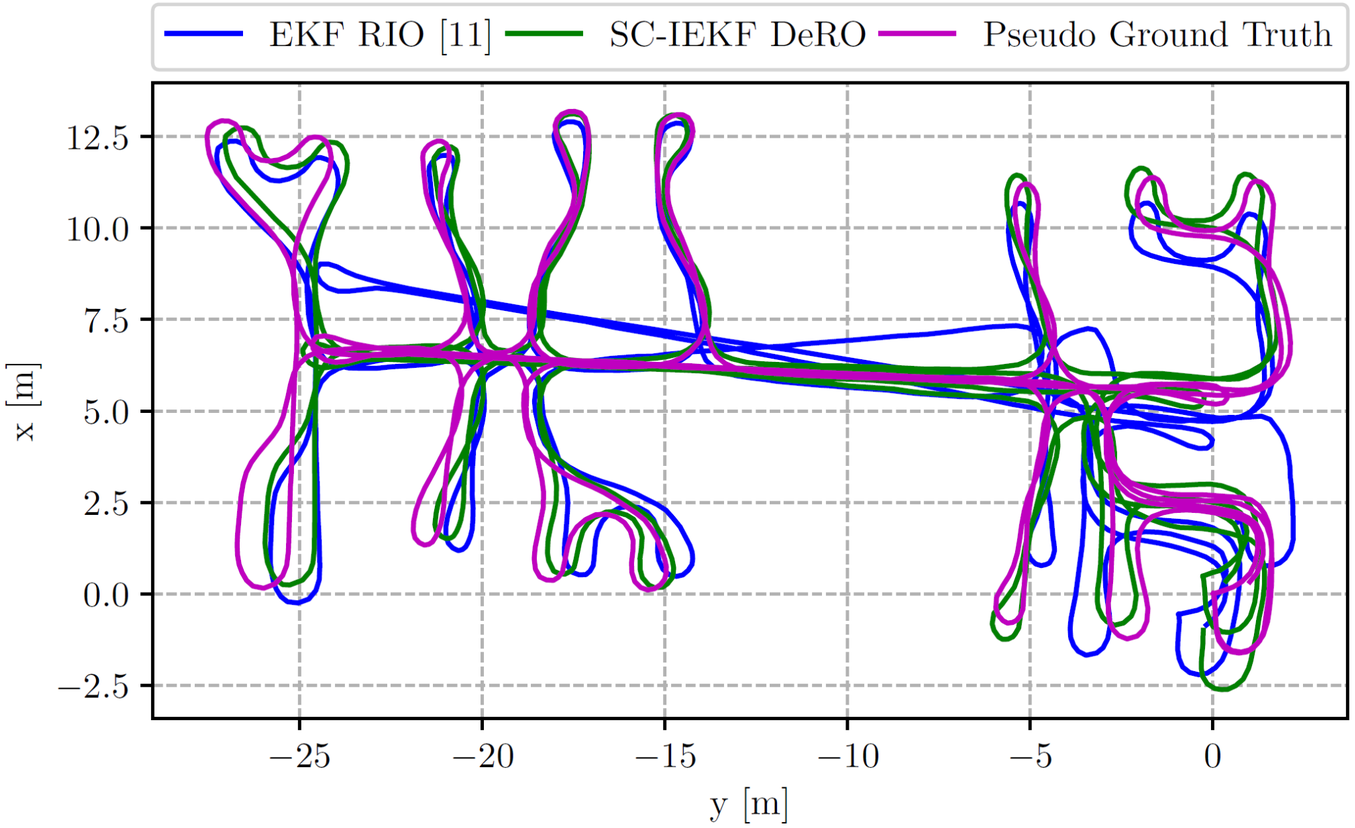


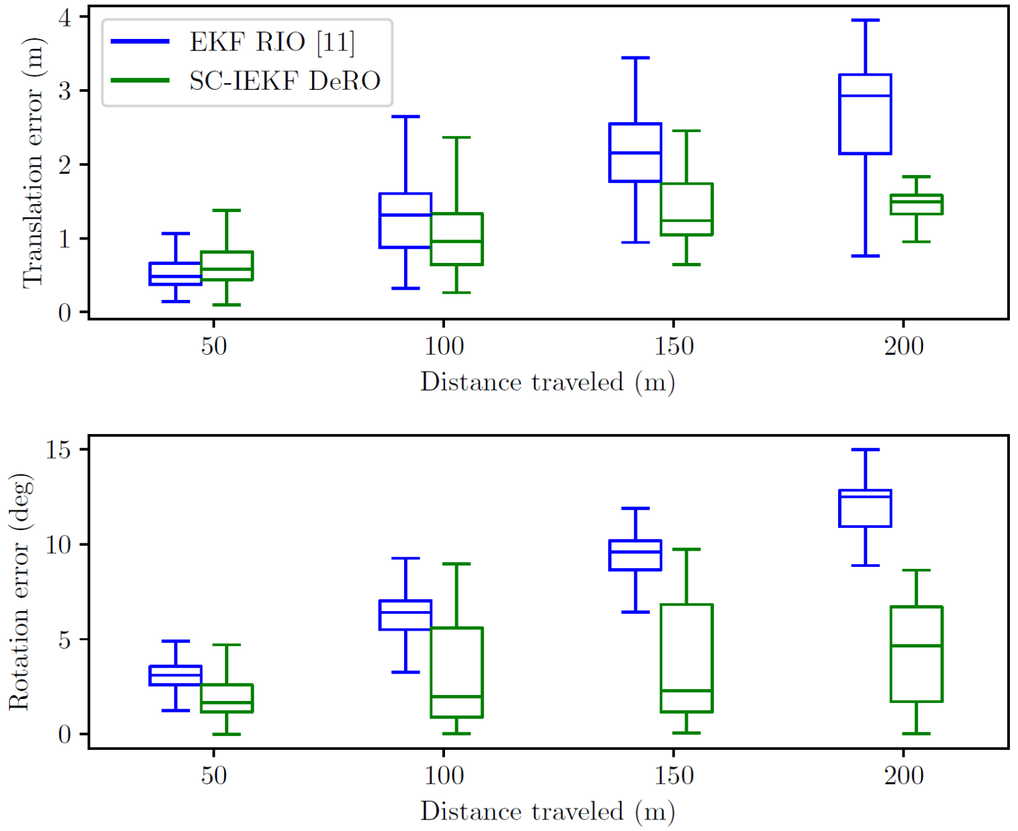


DeRO on GitHub

Experiments (3/6)

- For evaluation, we use a widely-used toolbox.
  - The estimated trajectory is aligned with the *position-yaw* option.
  - Metric: Absolute Trajectory Error (ATE).





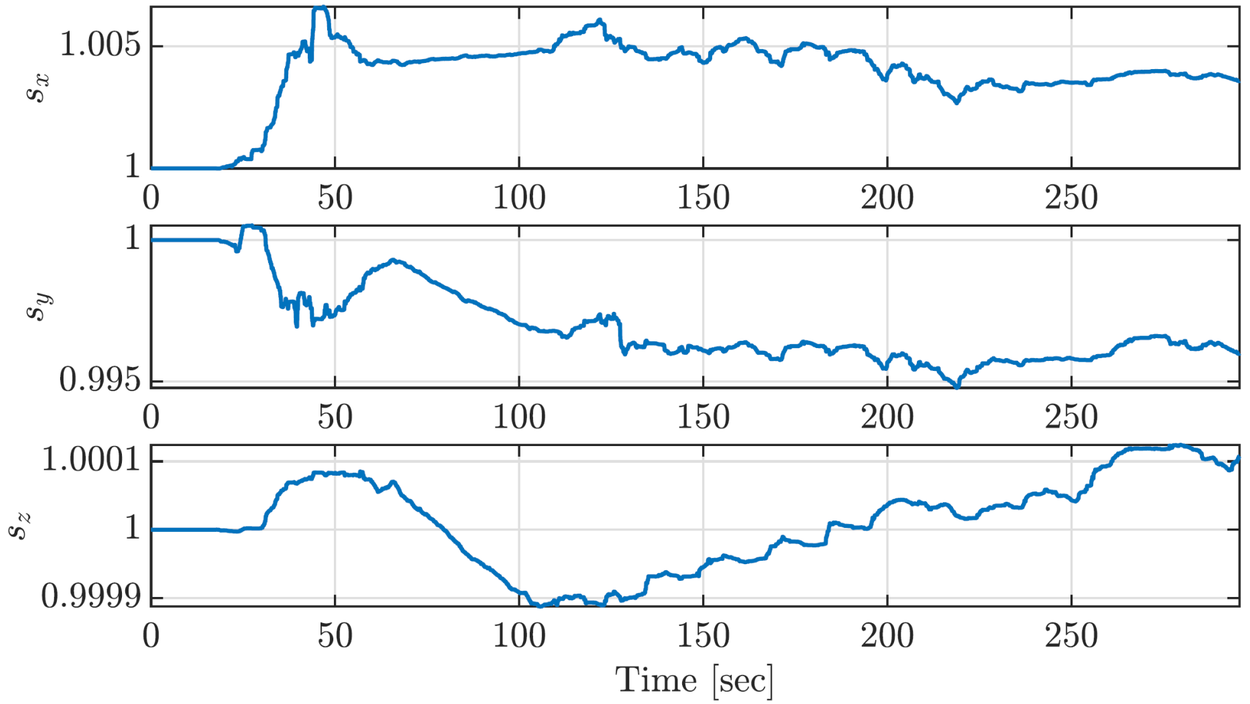
Z. Zhang and D. Scaramuzza, “A tutorial on quantitative trajectory evaluation for visual(-inertial) odometry,” in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2018, pp. 7244–7251.

Experiments (4/6)

- Overall, DeRO outperforms RIO.
  - DeRO reduces the translation error by approximately 62% and the rotation error by 66%.

Datasets	EKF RIO [11]		SC-IEKF DeRO	
	[m]	[deg]	[m]	[deg]
Carried 1 (287 m, 273 s)	1.075	5.370	**0.617**	**2.734**
Carried 2 (451 m, 392 s)	2.724	14.768	**0.709**	**2.869**
Carried 3 (235 m, 171 s)	1.110	5.741	**0.464**	**1.853**
Carried 4 (311 m, 220 s)	1.547	7.448	**0.491**	**2.659**
Carried 5 (228 m, 172 s)	1.495	5.331	**0.741**	**3.093**
Mean ATE	1.590	7.731	**0.604**	**2.641**

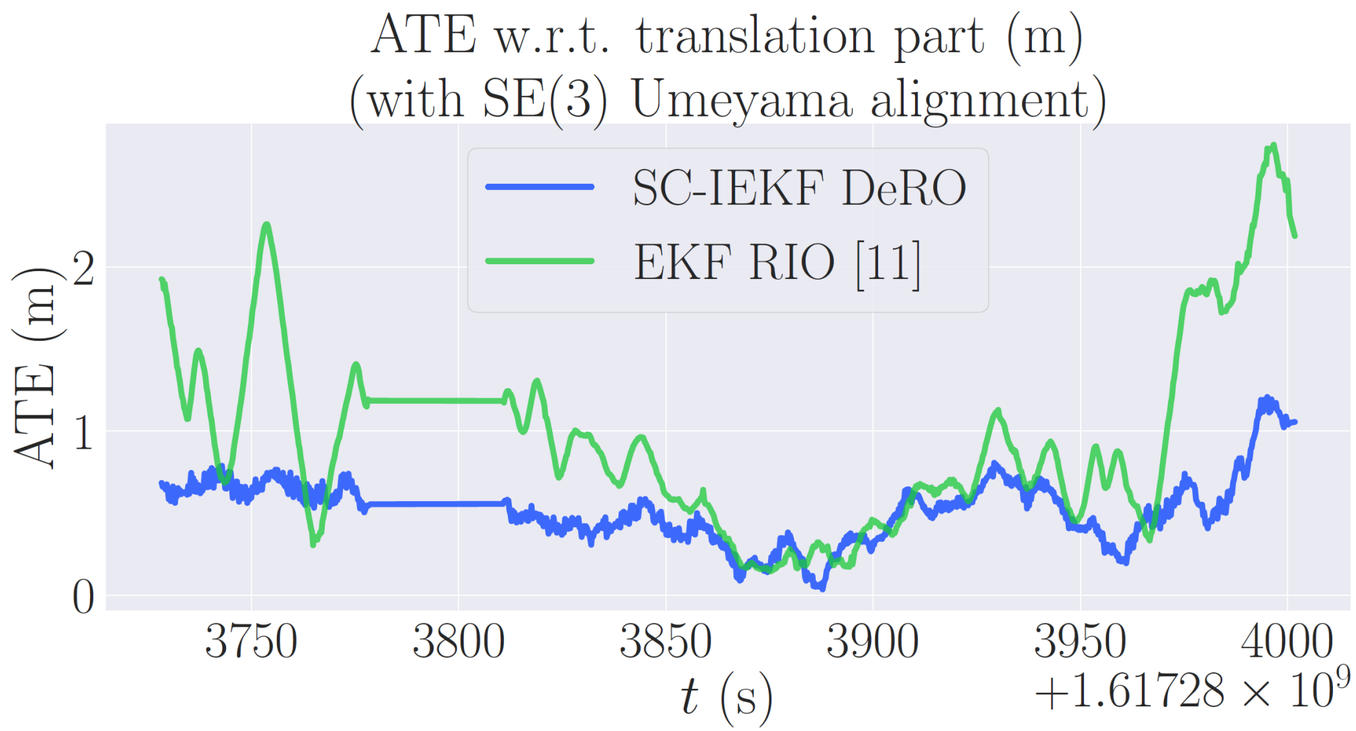
The bold numbers represent better results (the smaller number).

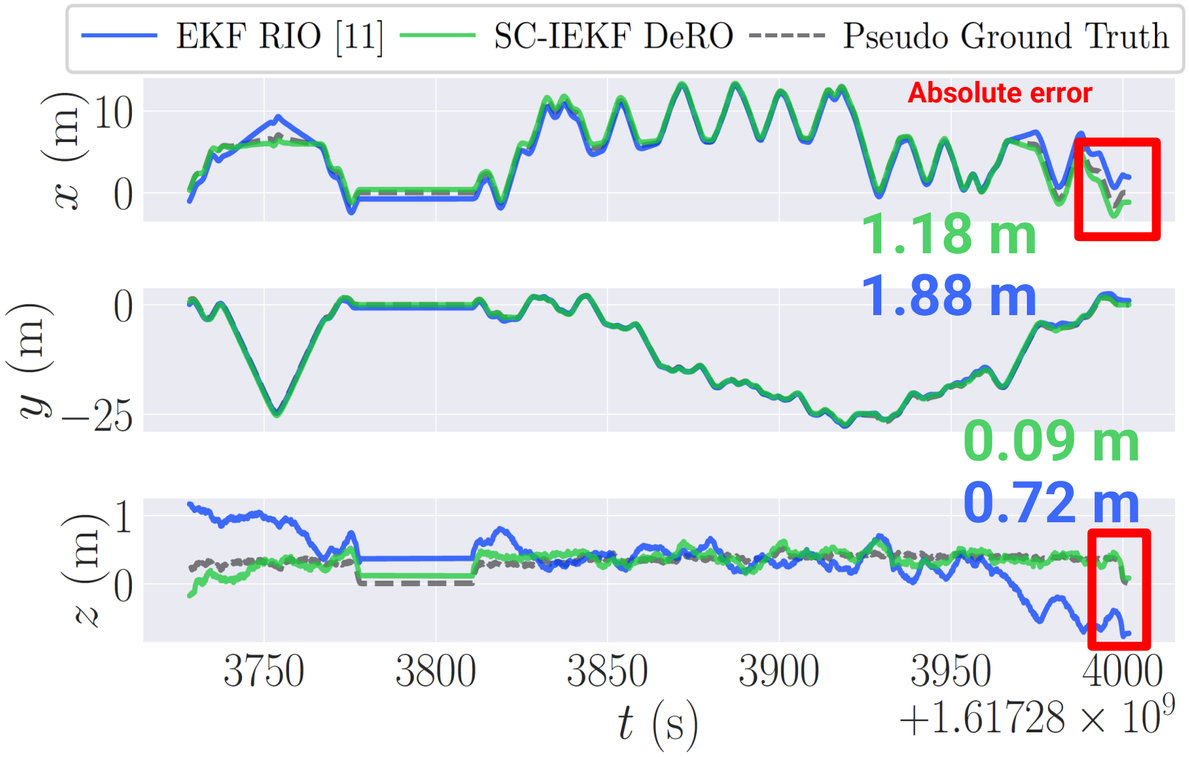


[11] C. Doer and G. F. Trommer, “An ekf based approach to radar inertial odometry,” in *2020 IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI)*, 2020, pp. 152–159.

Experiments (5/6)

- Additionally, we perform evaluation using another toolbox.
  - DeRO significantly improves the accuracy of z-axis position estimation.

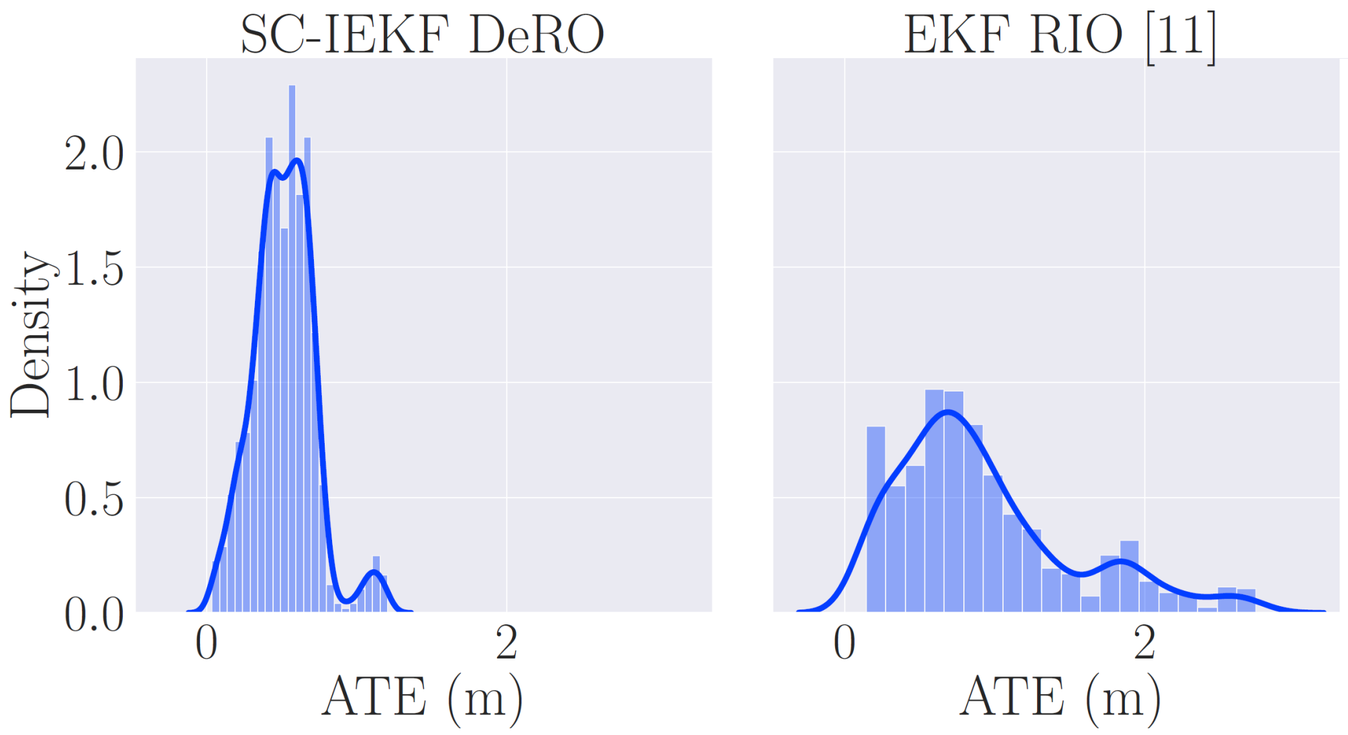


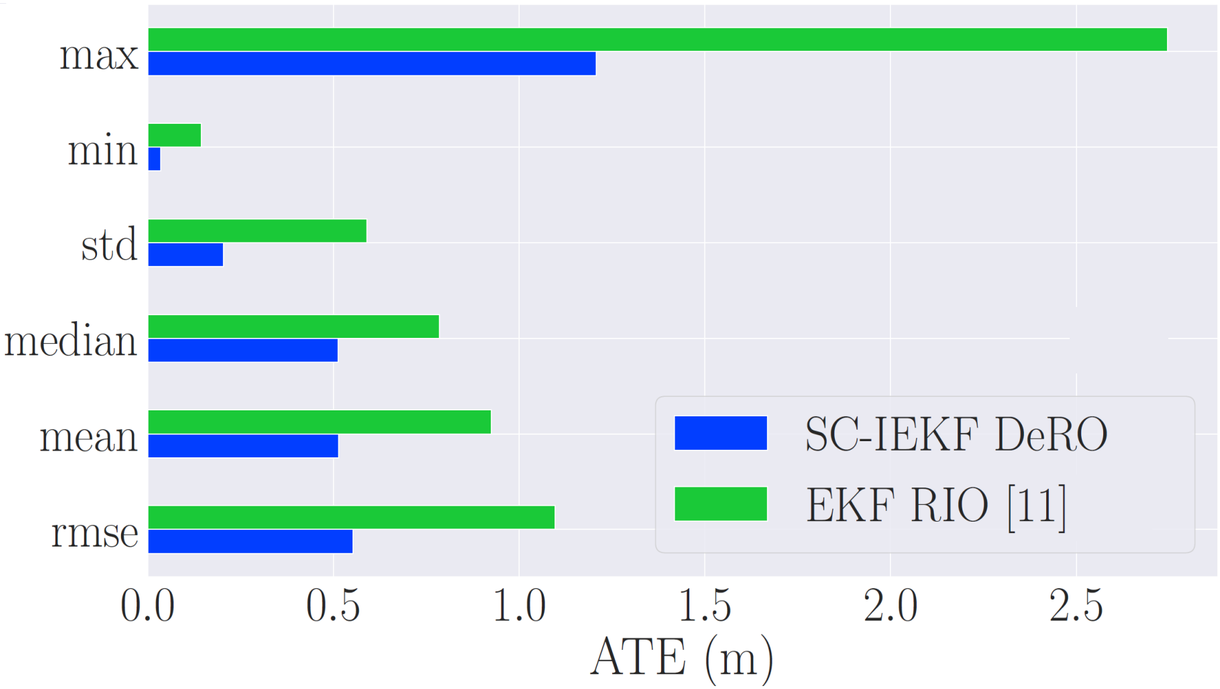


M. Grupp, “evo: Python package for the evaluation of odometry and slam.” https://github.com/MichaelGrupp/evo, 2017.

Experiments (6/6)

- DeRO yields a smaller ATE across all categories.





Conclusion

- We have proposed DeRO, a framework of dead reckoning based on radar odometry with tilt angle measurements.
- Our Approach
  - Avoid double integration of accelerometer in computing odometry.
  - Compensate Doppler velocity scale factor error.
- Rigorously evaluation using open-source dataset and toolbox.
- Make our source code openly available for the benefit of the research community.

Thank you!



Question & Answer

Feel free to check our DeRO on GitHub repository!